

Numerical Analysis Exam – August, 2017

Do all **5** (five) problems

1. Let  $f(x) = e^x$ .
  - (a) Find the linear Taylor polynomial  $T_1(x)$  of  $f(x)$  expanded about  $x_0 = 1/2$  and give an estimate for the maximum of  $|T_1(x) - f(x)|$  on the interval  $[0, 1]$ .
  - (b) Find the linear minimax approximation  $P_1(x)$  to  $f(x)$  on  $[0, 1]$  and find the maximum error of  $|P_1(x) - f(x)|$  on  $[0, 1]$
  - (c) Find the linear least squares approximation to  $f$  on  $[0, 1]$ .

2. The midpoint method for DE's is

$$w_0 = \alpha$$

$$w_{n+1} = w_n + hf \left( t_n + \frac{h}{2}, w_n + \frac{h}{2}f(t_n, w_n) \right).$$

Apply this method to the IVP,  $y' = \lambda y$ ;  $y(0) = 1$ , with  $\lambda < 0$  and find the constant  $c$  so that  $|h\lambda| < c$  implies that  $w_n \rightarrow 0$  as  $n \rightarrow \infty$ .

3. Derive this three-point formula for the second derivative.

$$f''(x) = \frac{1}{h^2} (f(x_0 - h) - 2f(x_0) + f(x_0 + h)) - \frac{h^2}{12} f^{(4)}(\eta)$$

for some  $\eta \in [x_0 - h, x_0 + h]$ .

4. Let  $f(x) = 2^x$  Let  $x_0 = -1, x_1 = 0, x_2 = 1$ . Find the total error bound for the degree two interpolating polynomial  $p_2(x)$  with these nodes and  $f$  on the interval  $[-1, 1]$ , i.e. derive a  $K$  with

$$\max_{t \in [-1, 1]} |f(t) - p_2(t)| < K.$$

5. Let  $g \in C^2([a, b])$  and  $p \in (a, b)$  with  $g(p) = p, g'(p) = 0, g''(p) \neq 0$ .

- (a) Show there is an  $\epsilon > 0$  so that for all  $x \in [p - \epsilon, p + \epsilon]$ , we have  $g^n(x) \rightarrow p$  as  $n \rightarrow \infty$ .
- (b) With  $\epsilon$  as in part(a), show that for all  $x \in [p - \epsilon, p + \epsilon]$ , we have  $|g(x) - p| \leq M|x - p|^2$  where  $M = \max\{|g''(x)| : |x - p| \leq \epsilon\}/2$ .